Maharashtra State Board Of Technical Education, Mumbai

Learning and Assessment Scheme Of Diploma Courses for Working Professionals

Programme Name	: Diploma In Electrical Engineering (Working Professional)		
Programme Code	: PEE	**With Effect From Academic Year**	: 2024-25
Duration of Programme	: 6 Semester		
Semester	: Fourth	**Scheme**	: F

Sr No	Course Title	Abbrevation	Course Type	Course Code	Diploma Awarding Course	Total IKS Hrs for Sem.	Learning Scheme					Credits	Assessment Scheme											Total Marks
							Actual Contact Hrs./Week			Self Learning (Activity/ Assignment /Micro Project)	Notional Learning Hrs /Week		Paper Duration (hrs.)	Theory				Based on LL & TL				Based on Self Learning		
																		Practical						
							CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
														Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
(All Compulsory)																								
1	D.C. MACHINES AND TRANSFORMERS	DMT	DSC	314322	No	0	4	-	4	-	8	4	3	30	70	100	40	25	10	25#	10	-	-	150
2	FUNDAMENTALS OF POWER ELECTRONICS	FPE	SEC	313335	No	0	3	-	4	1	8	4	3	30	70	100	40	25	10	25@	10	25	10	175
3	ELECTRICAL MATERIAL AND WIRING PRACTICE	EMW	SEC	313015	No	0	1	-	4	1	6	3	-	-	-	-	-	50	20	25@	10	25	10	100
4	COMPUTER AIDED DRAWING AND SIMULATION	CDS	SEC	314008	No	0	-	-	4	-	4	2	-	-	-	-	-	25	10	25@	10	-	-	50
	Total					**0**	**8**	**0**	**16**	**2**		**13**		**60**	**140**	**200**		**125**		**100**		**50**		**475**

Abbreviations : CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, FA - Formative Assessment,SA -Summative Assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
Legends : @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination
Note :
1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

Course Category : Discipline Specific Course Core (DSC) , Discipline Specific Elective (DSE) , Value Education Course (VEC) , Intern./Apprenti./Project./Community (INP) , AbilityEnhancement Course (AEC) , Skill Enhancement Course (SEC) , GenericElective (GE)

Programme Name/s	**: Electrical Engineering/ Electrical and Electronics Engineering/ Electrical Power System**
Programme Code	**: EE/ EK/ EP**
Semester	**: Fourth**
Course Title	**: D.C. MACHINES AND TRANSFORMERS**
Course Code	**: 314322**

I. RATIONALE

Despite advancements in electrical technology, D.C. machines still find applications in various industries and commercial sectors. Further the Transformers are essential components of power systems. This course is to equip students with fundamental knowledge, practical skills and a strong foundation in electrical power system and related fields.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Maintain D.C. Machines and Transformers used in Industry and related field.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Test the performance of D.C. Generators.
- CO2 - Test the performance of D.C. Motors
- CO3 - Test the performance of Single phase transformers
- CO4 - Use three phase transformer for different applications.
- CO5 - Use relevant special purpose transformers for different applications.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

D.C. MACHINES AND TRANSFORMERS **Course Code : 314322**

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
314322	D.C. MACHINES AND TRANSFORMERS	DMT	DSC	4	-	4	-	8	4	3	30	70	100	40	25	10	25#	10	-	-	150

Total IKS Hrs for Sem. : 0 Hrs
Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment
Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Describe the constructional details of the D.C. Machine. TLO 1.2 Explain the principle of working of the given D.C. Generator. TLO 1.3 Derive EMF equation and calculate the parameters of the given D.C. Generator. TLO 1.4 Identify the given type of D.C. Generator. TLO 1.5 Interpret the characteristics of the given D.C. Generator.	**Unit - I D.C. Generator** 1.1 D.C. Machine: construction, parts-function and material, types of winding (lap and wave) 1.2 D.C. Generator: Principle of operation, Faraday's law of electromagnetic induction, Fleming's right hand rule. 1.3 E. M. F. equation of D.C. Generator (derivation) 1.4 Types of D.C. Generator and it's applications. 1.5 Characteristics -internal and external.	Chalk-Board Flipped Classroom Video Demonstrations Model Demonstration Presentations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Explain the working principle of D.C. Motor. TLO 2.2 Apply the back emf equation in the given situation. TLO 2.3 Select relevant D.C. Motor for given application with justification. TLO 2.4 Calculate the torque, speed, output power and efficiency of the given D.C. Motor. TLO 2.5 Explain the various speed control methods of the given D.C. Motor. TLO 2.6 Describe with sketch the working of the starter for the given type of D.C. Motor. TLO 2.7 Describe the procedure of testing a D.C. Motor for the given condition. TLO 2.8 Explain with a diagram the working of the brushless D.C. Motor.	**Unit - II D.C. Motor** 2.1 D.C. Motor: Principle of operation, Lorentz force, Fleming's Left hand rule, Back emf and it's significance, Armature reaction. 2.2 Types of D.C. Motors, Torque: armature torque, shaft torque, Break Horse Power (BHP). 2.3 D. C. Motor characteristics- speed-armature current, torque- armature current, speed-torque. 2.4 Speed control: D.C. shunt and series motor- flux and armature control. 2.5 Starters, necessity of starters, two-point starters, three-point starters and four-point starters. 2.6 Testing: Break load test, Different types of losses, efficiency. 2.7 D.C. Motors applications, advantages and disadvantages 2.8 Brushless D.C. Motor: construction working, applications, advantages and disadvantages	Chalk-Board, Flipped Classroom Video, Demonstrations Model, Demonstration Presentations.

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Describe the constructional details of the single-phase transformer. TLO 3.2 Explain the working principle of single-phase transformer. TLO 3.3 Derive the EMF equation of transformer and calculate parameter for the given situations. TLO 3.4 Identify the type of single-phase transformer based on the given criterion. TLO 3.5 Interpret the name plate rating of the given transformer. TLO 3.6 Explain phasor diagram for no load/on load condition for the given type of transformer. TLO 3.7 Calculate regulation and efficiency by O.C. / S.C. tests and direct loading for the given type of transformer.	**Unit - III Single Phase Transformer** 3.1 Single phase transformer: Introduction, construction, parts-functions and material. 3.2 Principle of operation, EMF equation, voltage transformation ratio, turns ratio. 3.3 Types and losses, significance of transformer ratings. 3.4 No-load and On-load test on transformer and it's phasor diagram, Leakage reactance. 3.5 Equivalent circuit of transformer with equivalent resistances and reactances. 3.6 Voltage regulation and Efficiency: Direct loading. O.C. / S.C. method. All day efficiency, applications.	Chalk-Board, Flipped Classroom Video, Demonstrations Model, Demonstration Presentations.

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Describe the constructional details of the three - phase transformer. TLO 4.2 Identify the given type of transformer. TLO 4.3 Describe with diagrams various connections of the given three phase transformers. TLO 4.4 Select appropriate transformer on the given application. TLO 4.5 Describe the requirements for the parallel operation of the transformer. TLO 4.6 Describe the procedure for the given type of test on three phase transformer. TLO 4.7 Explain the importance of 'K' factor of transformers.	**Unit - IV Three Phase Transformer** 4.1 Three phase transformer: Introduction, construction, bank of three single phase transformers. Single unit of three phase transformer. 4.2 Working principle of three phase transformer. Types of transformers. 4.3 Connections as per IS: 2026 (part IV)-1977. Three phase to two phase conversion (Scott Connection). 4.4 Selection criteria as per IS: 10028 (Part I)-1985 of distribution transformer and power transformer, amorphous core type distribution transformer, specifications of three-phase distribution transformers as per IS:1180 (part I)-1989. 4.5 Need of parallel operation, conditions for parallel operation. 4.6 Polarity tests on mutually inductive coils, Phasing out test on Three- phase transformer. 4.7 Harmonics and their effects on transformer operation. 4.8 'K' factor of transformers: overheating due to non-linear loads and harmonics.	Chalk-Board Flipped Classroom Video, Demonstrations Model, Demonstration Presentations.

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Describe the constructional details of the given type of special purpose transformer. TLO 5.2 Explain the Working principle of the given type of special purpose transformer. TLO 5.3 State the applications of the given type of special purpose transformer.	**Unit - V Special Purpose Transformer** 5.1 Auto transformer: Construction, working and applications for single and three phases. 5.2 Instrument Transformers: Construction, working and applications of current transformer and potential transformer. 5.3 Isolation transformer: Construction, features and applications. 5.4 Single phase welding transformer: Construction, features and applications. 5.5 Pulse transformer: Construction, features and applications.	Chalk-Board, Flipped Classroom Video, Demonstrations Model, Demonstration Presentations.

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Identify different parts of the D.C. machine.	1	*Dismantling of a D.C. machine.	2	CO1
LLO 2.1 Verify generated output of the D.C. Shunt Generator.	2	*Measurement of D.C. Shunt Generator voltage by changing flux and speed.	2	CO1
LLO 3.1 Test the performance of D.C. Shunt generator.	3	*Load test on D.C. Shunt Generator.	2	CO1

D.C. MACHINES AND TRANSFORMERS **Course Code : 314322**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 4.1 Test the performance of D.C. Compound generator.	4	Load test on D.C. Compund Generator.	2	CO1
LLO 5.1 Test the performance of D.C. Shunt generator by Hopkinson's Test .	5	Testing the performance of D.C. Shunt generator by Hopkinson's Test .	2	CO1
LLO 6.1 Reverse the direction of rotation of the D.C. shunt motor.	6	*Reversal of rotation of D.C. shunt motor.	2	CO2
LLO 7.1 Perform brake test on D.C. shunt motor.	7	*Speed torque characteristics of D.C. shunt motor.	2	CO2
LLO 8.1 Control the speed of D.C. shunt motor by different methods.	8	*Speed control of D.C. shunt motor using Armature control & flux control method.	2	CO2
LLO 9.1 Reverse the direction of rotation of the D.C. series motor.	9	*Reversal of rotation of D.C. series motor.	2	CO2
LLO 10.1 Control the speed of the D.C. series motor by different methods.	10	Speed control of D.C. series motor using Armature control & flux control method.	2	CO2
LLO 11.1 Perform brake test on a D.C. series motor.	11	Brake test on D.C. series motor.	2	CO2
LLO 12.1 Reverse the direction of rotation of the D.C. compound motor.	12	*Reversal of rotation of D.C. compound motor.	2	CO2
LLO 13.1 Identify different parts of a three point starter of a D.C. Shunt Motor. LLO 13.2 Check the function of the various parts of three point starter.	13	*Demonstration of operating mechanism of three point starter of a D.C. Shunt Machine.	2	CO2

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 14.1 Identify different parts of a four point starter of a D.C. Compound Motor. LLO 14.2 Check the function of the various parts of four point starter.	14	*Demonstration of operating mechanism of four point starter of a D.C. Compound Machine.	2	CO2
LLO 15.1 Identify different parts of a two point starter of a DC series Motor. LLO 15.2 Check the function of the various parts of two point starter.	15	*Demonstration of operating mechanism of two point starter of a DC series Machine.	2	CO2
LLO 16.1 Identify the different parts of single phase & Three phase transformer.	16	*Demonstration of a single phase & Three phase transformer construction.	2	CO3 CO4
LLO 17.1 Find the transformation ratio of single phase transformer.	17	* Transformation ratio of single phase transformer.	2	CO3
LLO 18.1 Test the performance of single phase transformer.	18	*Direct load test of single phase transformer.	2	CO3
LLO 19.1 Test the performance of single phase transformer.	19	*Open circuit and short circuit test on single phase transformer to determine equivalent circuit parameters.	2	CO3
LLO 20.1 Test the performance of single phase transformer.	20	*Open circuit and short circuit test on single phase transformer to determine voltage regulation and efficiency.	2	CO3
LLO 21.1 Perform parallel operation of two single phase transformers.	21	*Perform parallel operation of two single phase transformers to determine the load current sharing.	2	CO3

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 22.1 Perform parallel operation of two single phase transformers.	22	*Perform parallel operation of two single phase transformers to determine the apparent and real power load sharing.	2	CO3
LLO 23.1 Perform polarity test on a single phase transformer.	23	*Perform polarity test on a single phase transformer whose polarity markings are masked.	2	CO3
LLO 24.1 Convert three phase to two phase conversion by Scott-Connection.	24	Scott-Connection of three phase transformer.	2	CO4
LLO 25.1 Perform Back to Back test on single phase transformer.	25	*Back to Back test on single phase transformer.	2	CO4
LLO 26.1 Connect the auto-transformer in step-up and step-down modes, measure input and output voltage.	26	*Connection of the auto-transformer.	2	CO5
LLO 27.1 Verify the Current transformer (CT) ratio.	27	Functioning of the Current transformer (CT).	2	CO5
LLO 28.1 Verify the Potential Transformer (PT) ratio.	28	Functioning of the Potential Transformer (PT).	2	CO5
LLO 29.1 Verify turns ratio of the isolation transformer.	29	*Functioning of the isolation transformer.	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

NO SLA

- Not applicable for this course

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	DC series and shunt machines (up to 230 V, 4 kW).	1,2,3,4,5,6,7,8,9,10,11,12,13,14,15
2	Three point starter.	13
3	Four point starter.	14

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
4	Two point starter.	15
5	Single phase transformer of suitable size (500 VA to 1 kVA).	16,17,18,19,20,21,22,23
6	Three phase transformer of suitable size (1 kVA to 3 kVA).	16,24,25
7	AC Ammeter range (0-2.5-5-10A). Portable analog MI type as per relevant BIS standard.	17,18,19,20,21,22,23,24,25,26,27,28,29
8	AC Voltmeter Range (0-75/150/300V), Portable analog MI type as per relevant BIS standard.	17,18,19,20,21,22,23,24,25,26,27,28,29
9	Wattmeter 0-300/600 V, 5/10 A, for use in A.C. circuits.	18,19,20,22
10	L.P.F. Wattmeter, 0-300/600 V, 1A to 2A, for use in A.C. circuits.	19,20
11	Lamp load of 10-20 A.	2,3,4,5,13,14,15,17,18,21,22,29
12	DC Supply, 230 V, 25 A.	2,3,4,5,6,7,8,9,10,11,12,13,14,15,23
13	Rheostat (0-500 Ohm. 1.2A), Nichrome wire wound rheostat on epoxy resin or class F insulating tube with two fixed and one sliding contact.	2,3,4,5,7,8,10,11
14	Tachometer(0-10,000 RPM).	2,3,4,5,7,8,10,11
15	DC Ammeter range (0-5-10A), Portable analog PMMC type as per relevant BIS standard.	2,3,4,5,7,8,10,11,12
16	DC Voltmeter Range (0-150/300V), Portable analog PMMC type as per relevant BIS standard.	2,3,4,5,7,8,10,11,12
17	Rheostat (0-400 Ohm, 1.5A). Nichrome wire wound rheostat on epoxy resin or class F insulating tube with two fixed and one sliding contact.	2,5,7,8,10,11
18	Single phase auto transformer 0-270 V, 15 A.	26
19	CT of suitable ratio.	27
20	PT of suitable ratio.	28
21	Isolation transformer of suitable ratio.	29

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
22	Rheostat (0-100 Ohm, 5A), Nichrome wire wound rheostat on epoxy resin or class F insulating tube with two fixed and one sliding contact.	7,8,10
23	Rheostat (0-50 Ohm, 10A), Nichrome wire wound rheostat on epoxy resin or class F insulating tube with two fixed and one sliding contact.	7,8,10,11

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	D.C. Generator	CO1	8	2	0	6	8
2	II	D.C. Motor	CO2	16	6	4	10	20
3	III	Single Phase Transformer	CO3	17	2	8	10	20
4	IV	Three Phase Transformer	CO4	13	2	8	6	16
5	V	Special Purpose Transformer	CO5	6	2	0	4	6
Grand Total				**60**	**14**	**20**	**36**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Two unit tests of 30 marks will be conducted and an average of two unit tests considered.
- For formative assessment of laboratory learning 25 marks.
- Each practical will be assessed considering appropriate % weightage to process and product and other instructions of assessment.

Summative Assessment (Assessment of Learning)

- End semester summative assessment of 25 marks for laboratory learning.
- End semester assessment of 70 marks through offline mode of examination.

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3	1	1	2	1	1	1			
CO2	3	2	2	2	3	1	2			
CO3	3	1	1	2	2	1	1			
CO4	3	2	2	2	3	1	1			
CO5	3	1	1	2	2	1	1			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number

Sr.No	Author	Title	Publisher with ISBN Number
1	Bhattacharya,S. K	Electrical Machines	McGraw Hill Education, New Delhi ISBN-13:978-0070669215
2	Mehta, V. K. and Mehta, Rohit	Principles of Electrical Machines	S.Chand and Co.Ltd., New Delhi ISBN-13: 978-8121921916
3	Theraja B. L.	Electrical Technology Vol-II (AC and DC machines)	S.Chand and Co.Ltd., New Delhi ISBN-13: 978-8121924375
4	Bandyopadhyay M. N.	Electrical Machines Theory and Practice	PHI Learning Pvt. Ltd., New Delhi ISBN-13:978-8120329973
5	Mittle, V.N. and Mittle, Arvind	Basic Electrical Engineering	McGraw Hill Education, New Delhi ISBN-13: 978-0070593572
6	Kothari, D. P. and Nagrath, I. J.	Electrical Machines	McGraw Hill Education, New Delhi ISBN-13:978-9352606405
7	Murugesh Kumar K.	DC Machines and Transformers	S. Chand, ISBN-13: 978-8125916055
8	J. B. Gupta	Theory & Performance of Electrical Machine	S-K-Kataria, ISBN-13: 978-9350142776

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://youtu.be/D4RFFnzRdkk?si=d5iNRWSZbl01NvT3	Construction & Working Principle of a D.C. Machine.
2	https://youtu.be/1OfLgpFq6Rc?si=bwN9d7ESlV2Utzb6	D.C. Motors.
3	https://youtu.be/6dF3LDzb-tE?si=OYZMdgs2I5d7bqAa	D.C. Generator.

Sr.No	Link / Portal	Description
4	https://youtu.be/qmcriUdYBW0?si=ea5Sa1G9R9m7aRTm	Transformer.
5	www.nptel.ac.in	About construction, working principle and operation of D.C. Machine, single phase transformer, three phase transformer and special purpose transformer.
6	www.electricaltechnology.org	About construction, working principle and operation of D.C. Machine, single phase transformer, three phase transformer and special purpose transformer.
7	www.electrical4u.com	About construction, working principle and operation of D.C. Machine, single phase transformer, three phase transformer and special purpose transformer.

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

Programme Name/s	**: Electrical Engineering/ Electrical Power System**
Programme Code	**: EE/ EP**
Semester	**: Third**
Course Title	**: FUNDAMENTALS OF POWER ELECTRONICS**
Course Code	**: 313335**

I. RATIONALE

Power Electronics finds extensive applications in domestic, commercial, industrial front and electric utilities particularly in terms of efficient conversion, control and conditioning of electric power from its available input into the desired electrical output form. This course will enable the diploma students to develop the knowledge and skill sets of operating and testing different power electronic devices and their applications.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Test the Performance of Power Electronic Devices and Circuits.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Test the functionality of a given power electronic device.
- CO2 - Test the switching performance of a thyristor.
- CO3 - Test the performance of given controlled converter.
- CO4 - Test the performance of given chopper.
- CO5 - Use suitable power electronic circuit for given application.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

FUNDAMENTALS OF POWER ELECTRONICS Course Code : 313335

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											Total Marks
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
313335	FUNDAMENTALS OF POWER ELECTRONICS	FPE	SEC	3	-	4	1	8	4	3	30	70	100	40	25	10	25@	10	25	10	175

Total IKS Hrs for Sem. : 0 Hrs

Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

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5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Illustrate the power electronic system using block diagram. TLO 1.2 Solve simple numerical on losses in the given switch. TLO 1.3 Explain the general characteristics of the given power electronic switch. TLO 1.4 Describe the construction of the given power electronic device. TLO 1.5 Explain the working principle of the given power electronic device. TLO 1.6 State the applications of the given power electronic device.	**Unit - I Power Electronic Devices** 1.1 Power electronic system: general block diagram, need, advantages and disadvantages 1.2 Switching in power electronic circuit: Need and its importance; Ideal switch and practical switch: concept, general characteristics, conduction losses, switching losses 1.3 SCR: Construction, working principle, Static V-I characteristics, switching characteristics, and applications 1.4 IGBT: Construction, working principle, Static V-I characteristics, switching characteristics, and applications 1.5 Power MOSFET: Construction, working principle, Static V-I characteristics, and applications 1.6 TRIAC: Construction, working principle, Static V-I characteristics, and applications	Lecture Using Chalk-Board Demonstration Video presentations Flipped Classroom

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Explain the need of the given protection for SCR TLO 2.2 Describe the given protection scheme of SCR TLO 2.3 Explain the given turn-on method of SCR TLO 2.4 Illustrate the given firing circuit of SCR TLO 2.5 Explain the given commutation technique of SCR	**Unit - II Protection and Firing Circuit of Thyristor** 2.1 di/dt protection: need, snubber circuit 2.2 dv/dt protection: need, snubber circuit 2.3 Overvoltage protection: need, internal & external overvoltage, voltage clamping device 2.4 Overcurrent protection: need, electronic crowbar circuit 2.5 Thermal Protection of SCR: Need, thermal resistance, and heat sink specification 2.6 Firing circuit: Features and general layout of firing scheme 2.7 SCR turn-on methods: forward voltage triggering, gate triggering, dv/dt triggering, temperature triggering, and light triggering 2.8 SCR Firing circuit: resistance firing circuit (no derivation), RC firing circuit (no derivation), pulse transformer based triggering 2.9 SCR commutation techniques: load commutation (Class A), line commutation (Class F)	Lecture Using Chalk-Board Presentations Video Demonstrations Flipped Classroom

FUNDAMENTALS OF POWER ELECTRONICS **Course Code : 313335**

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Define the given term(s) related to controlled converter. TLO 3.2 Illustrate the working of the given single phase controlled rectifier. TLO 3.3 Derive equation of DC output voltage of the given controlled converter. TLO 3.4 Compare voltage source inverter and current source inverter on the basis of the given criteria. TLO 3.5 Explain working of the given single phase inverter. TLO 3.6 Explain working principle of sinusoidal pulse width modulation.	**Unit - III Controlled Converters** 3.1 Basic terminologies: conduction angle, firing angle, output voltage, output current, voltage across switch, source current, source voltage 3.2 Single phase half wave controlled rectifier with R, RL load: Circuit diagram, working, input-output waveforms, derivation for average output voltage, equations for output currents, voltages & power, and effect of freewheeling diode 3.3 Single phase full wave controlled bridge rectifier with R, RL load: Circuit diagram, working, input-output waveforms, derivation for average output voltage, equations for output currents, voltages & power 3.4 Three phase full wave controlled bridge rectifier: working principle with R load, input-output waveforms 3.5 Inverters: concept of voltage source inverter and current source inverter 3.6 Single phase half wave bridge inverter with R, RL load: Circuit diagram, working, input-output waveforms 3.7 Single phase full wave bridge inverter with R, RL load: Circuit diagram, working, input-output waveforms 3.8 Pulse width modulation: importance/need, types; Sinusoidal pulse width modulation: concept, working principle and waveforms	Lecture Using Chalk-Board Presentations Video Demonstrations Flipped Classroom

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Explain the given terminology related to chopper. TLO 4.2 Explain control strategies of chopper. TLO 4.3 Illustrate working of the given chopper. TLO 4.4 Calculate output voltage of the given chopper.	**Unit - IV DC-DC Converters** 4.1 Basic terminologies: duty ratio, turn on period, turn off period, chopping period 4.2 Control strategies of chopper: Constant frequency system, variable frequency system 4.3 Step up chopper: circuit diagram, working, waveforms and output voltage equation 4.4 Step down chopper: circuit diagram, working, waveforms and output voltage equation 4.5 Buck-Boost chopper: circuit diagram, working, waveforms and output voltage equation	Lecture Using Chalk-Board Presentations Video Demonstrations Flipped Classroom
5	TLO 5.1 Explain the operation of charge controller used in the photovoltaics (PV) system. TLO 5.2 Explain speed control of ceiling fan using TRIAC. TLO 5.3 Explain AC to AC converter used in Wind Power Generation. TLO 5.4 Explain the function of converter station in HVDC.	**Unit - V Applications of Power Electronics** 5.1 Charge Controller: Concept, types, applications in Photovoltaics (PV) system with block diagram 5.2 Speed control of ceiling fan using TRIAC: Working, Block Diagram, advantages 5.3 AC to AC converter using DC link: Concept, applications in Wind Power Generation 5.4 HVDC converter station: Concept, Circuit Diagram	Lecture Using Chalk-Board Presentations Video Demonstrations Site/Industry Visit Case Study

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Identify given power electronic device	1	*Power Electronic Devices.	2	CO1
LLO 2.1 Test the performance of SCR.	2	*V-I Characteristics of SCR.	2	CO1
LLO 3.1 Test the proper functioning of the power MOSFET.	3	*Testing of power MOSFET	2	CO1
LLO 4.1 Test the proper functioning of the IGBT.	4	*Testing of IGBT	2	CO1
LLO 5.1 Test the proper functioning of the TRIAC.	5	Testing of TRIAC.	2	CO1
LLO 6.1 Test the performance of Snubber circuit.	6	*Performance of Snubber circuit.	2	CO2
LLO 7.1 Test the effect of variation of resistance in R triggering circuit on the firing angle of SCR.	7	*Resistance triggering circuit of SCR.	2	CO2
LLO 8.1 Test the effect of variation of resistance and capacitance in RC triggering circuit on the firing angle of SCR.	8	RC triggering circuit of SCR	2	CO2
LLO 9.1 Perform the triggering of SCR using Pulse transformer	9	Triggering of SCR using Pulse transformer	2	CO2
LLO 10.1 Perform the operation of Class A commutation circuit.	10	*Class A (Load Commutation) commutation circuit.	2	CO2
LLO 11.1 Perform the operation of Class F commutation circuit.	11	Class F (Line Commutation) commutation circuit	2	CO2

FUNDAMENTALS OF POWER ELECTRONICS **Course Code : 313335**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 12.1 Measure output voltage of single phase half wave controlled rectifier by using CRO/DSO. LLO 12.2 Use various controls of the CRO/DSO	12	*Operation of single phase half wave controlled rectifier with resistive load.	2	CO2 CO3
LLO 13.1 Measure output voltage of single phase half wave controlled rectifier by using CRO/DSO. LLO 13.2 Use various controls of the CRO/DSO	13	Operation of single phase half wave controlled rectifier with RL load without freewheeling diode.	2	CO2 CO3
LLO 14.1 Measure output voltage of single phase half wave controlled rectifier by using CRO/DSO. LLO 14.2 Use various controls of the CRO/DSO	14	*Operation of single phase half wave controlled rectifier with RL load with freewheeling diode.	2	CO2 CO3
LLO 15.1 Measure output voltage of single phase full wave controlled rectifier by using CRO/DSO. LLO 15.2 Use various controls of the CRO/DSO	15	Operation of single phase full wave controlled rectifier with R load.	2	CO2 CO3
LLO 16.1 Measure output voltage of single phase full wave controlled rectifier by using CRO/DSO. LLO 16.2 Use various controls of the CRO/DSO	16	*Operation of single phase full wave controlled rectifier with RL load.	2	CO2 CO3
LLO 17.1 Measure output voltage of three phase full wave controlled rectifier by using CRO/DSO. LLO 17.2 Use various controls of the CRO/DSO	17	Operation of three phase full wave controlled rectifier with R load.	2	CO2 CO3

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 18.1 Measure the output voltage with different firing angles of the controlled rectifier available in your laboratory	18	*Voltage control using controlled rectifier.	2	CO2 CO3
LLO 19.1 Measure output voltage of single phase half wave bridge inverter by using CRO/DSO. LLO 19.2 Use various controls of the CRO/DSO	19	*Operation of single phase half wave bridge inverter with resistive load.	2	CO2 CO3
LLO 20.1 Measure output voltage of single phase full wave bridge inverter by using CRO/DSO. LLO 20.2 Use various controls of the CRO/DSO	20	Operation of single phase full wave bridge inverter with resistive load.	2	CO2 CO3
LLO 21.1 Measure output voltage of single phase half wave bridge inverter by using CRO/DSO. LLO 21.2 Use various controls of the CRO/DSO	21	Operation of single phase half wave bridge inverter with RL load.	2	CO2 CO3
LLO 22.1 Measure output voltage of single phase full wave bridge inverter by using CRO/DSO. LLO 22.2 Use various controls of the CRO/DSO	22	Operation of single phase full wave bridge inverter with RL load	2	CO2 CO3
LLO 23.1 Measure the output voltage of chopper by varying duty cycle. LLO 23.2 Use various controls of the CRO/DSO	23	Operation of step-up chopper.	2	CO2 CO4
LLO 24.1 Measure the output voltage of chopper by varying duty cycle. LLO 24.2 Use various controls of the CRO/DSO	24	*Operation of step-down chopper.	2	CO2 CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 25.1 Test the performance of charge controller in PV system.	25	Charge controller in PV system	2	CO4 CO5
LLO 26.1 Observe the operation of AC to AC converter (with DC link). LLO 26.2 Interpret the input and output profile of the AC to AC converter (with DC link).	26	*Demonstration of AC to AC converter (with DC link) used in wind power plant.	2	CO3 CO4 CO5
LLO 27.1 Control the speed of fan using TRIAC.	27	*Speed control of fan using TRIAC.	2	CO3 CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- Build a power electronic circuit to produce variable voltage for a given application using following steps. 1) Identify voltage range for a given application. 2) Select circuit components suitable for the identified voltage range. 3) Connect circuit components to build power electronic circuit controlling voltage. 4) Test the circuit for the production of variable voltage. 5) Prepare a report on the circuit built and submit the same.
- Prepare a report on commercial or industrial applications of power electronics devices by performing following activities. 1) Identify 3 to 5 relevant applications. 2) Visit site and understand role of power electronic devices in identified applications.

3) Write the specifications of major components in the applications. 4) Prepare block diagram or process flow diagram of the applications.

- Prepare a report on the ratings/specifications and applications of various power electronic devices. 1) Select any 3 to 5 power electronic devices. 2) Visit manufacturers' site or official websites of power electronic devices manufacturers and note the specifications or ratings of the selected power electronic devices. 3) Compare selected power electronics devices based on collected information along with their applications.
- Build a circuit of charge controller for a given battery using following steps. 1) Write specifications of a given battery. 2) Select circuit components required for charge controller circuit suitable for given battery. 3) Connect circuit components to build charge controller. 4) Test change controller for controlling power flow through battery. 5) Prepare a report on the charge controller and submit the same.
- Any other relevant microproject assigned by subject teacher.

Assignment

- Numerical on losses in power electronic device.
- Prepare a report on evolution of power electronic devices.
- Numerical on output voltage of given controlled converter.
- Numerical on DC output voltage of given chopper.
- Prepare a report on testing the performance of GTO.
- Any other relevant assignment given by subject teacher.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	SCR: Irms = 16A, IH = 100mA, IL = 200mA, IGT = 90 to 35 mA, VGT = 3 to 1 V, Vrms = 1600V	1,2,6,7,8,10,11,9,12,13,14,15,16,17,18
2	Power MOSFET: Vds-400V, ID-10A-6.3A Pd-125W	1,3,23,24
3	IGBT: Vces = 1200V, VGE = 20V, IC = 139 to 93A, PD = 650 to 300W	1,4,19,20,21,22
4	TRIAC: It = 4A, IGT = 10mA, Vt = 600V.	1,5
5	Rheostat: Nicrome wire, 300ohm, 10A, 400V	12,13,14,15,16,17,19,20,21,22
6	Variable inductive load: Single phase,250V, 2.5kW continuously variable core type	12,13,14,15,16,17,19,20,21,22
7	CRO/Digital Oscilloscope with probes: 20MHz, dual channel, sensitivity = 1mV/div., Max Input = 400V, Power supply = 230VAC.	12,13,14,15,16,17,19,20,21,22

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
8	Clamp on meter: Current = 0 to 400A, Voltage = 0 to 600V	2,3,4,5,25,27
9	AC and DC Ammeter: Range = 0 to 20A, Sensitivity = 0.5A/div.	25,27
10	AC and DC Voltmeter: 0 to 300V, Sensitivity = 1V/div.	25,27
11	Multimeter: 2000 count digital display, 1000V DC/750 V AC ranges, 10 A AC/DC ranges	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Power Electronic Devices	CO1	8	2	6	4	12
2	II	Protection and Firing Circuit of Thyristor	CO1,CO2	11	4	10	4	18
3	III	Controlled Converters	CO2,CO3	14	2	14	6	22
4	IV	DC-DC Converters	CO2,CO4	7	2	4	4	10
5	V	Applications of Power Electronics	CO3,CO4,CO5	5	2	4	2	8
Grand Total				**45**	**12**	**38**	**20**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- 30 Marks of Theory FA shall be obtained from an average mark of two unit tests (each of 30 marks) held in the semester. At least 2 COs should be covered in each unit test.
- Continuous assessment shall be based on process and product related performance indicators and laboratory experiences. Each practical shall be assessed for 25 marks considering 60% weightage to process and 40% weightage to product.

- Rubrics of continuous assessment of practical, including performance indicators, shall be designed by concerned course teacher.

Summative Assessment (Assessment of Learning)

- End semester, theory summative assessment of 70 marks shall be based on offline mode of written examination.
- End semester, practical summative assessment of 25 marks shall be based on student's performance in end semester practical exam.

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3	2	-	3	-	-	3			
CO2	3	2	1	3	-	-	2			
CO3	3	2	2	3	-	1	2			
CO4	3	2	2	3	-	1	2			
CO5	2	3	2	2	2	2	2			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Muhammad H. Rashid	Power Electronics Handbook	Butterworth-Heinemann Inc, ISBN:978-0128114070
2	P S. Bimbhra	Power Electronics	KHANNA PUBLISHERS, ISBN:978-8174092793
3	Muhammad H. Rashid	Power Electronics: Devices, Circuits, and Applications	Pearson Education, ISBN:978-8120345317
4	M D Singh, K B Khanchnadani	Power Electronics	McGraw Hill Education, ISBN:9780070583894

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://nptel.ac.in/courses/108102145	Course on Power Electronics by IIT Delhi
2	https://nptel.ac.in/courses/108105066	Course on Power Electronics by IIT Kharagpur
3	https://nptel.ac.in/courses/108101038	Course on Power Electronics by IIT Bombay
4	https://ocw.mit.edu/courses/6-334-power-electronics-spring-2007/	Course on Power Electronics by MIT Opencourseware
5	https://youtube.com/playlist?list=PLSnw1KE0TFkVu05Ws0Ax143gZYmxPMCoY&si=FWLw-jfnLxC_l-4j	Laboratory course on Power Electronics by RGUKT Basar

Sr.No	Link / Portal	Description
6	https://www.youtube.com/playlist?list=PL4emuJKx0B8aREwkC5BEOw2OZ48puPyOG	Videos on Power Electronics
7	https://3dcircuits.engineering.dartmouth.edu/powani.html	Animation on Chopper and Rectifier

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

Programme Name/s	**: Electrical Engineering/ Electrical Power System**
Programme Code	**: EE/ EP**
Semester	**: Third**
Course Title	**: ELECTRICAL MATERIAL AND WIRING PRACTICE**
Course Code	**: 313015**

I. RATIONALE

Electrical diploma engineers should be able to select relevant electrical materials and accessories for different applications while carrying out new work or maintenance work. They should be well conversant with the specifications of material as per the applications and wiring practices. This course will enable the students to identify and select the material for a particular application and also take up the wiring related work like, selection of material for wiring, carry out the wiring and testing etc.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Carry out wiring and maintenance activities using relevant materials, tools and safety practices.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Follow safe working practices when undertaking electrical work.
- CO2 - Select relevant conducting, electromagnetic and magnetic materials.
- CO3 - Select relevant insulating materials.
- CO4 - Perform different types of electrical wiring and cabling activities.
- CO5 - Implement relevant earthing systems.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

ELECTRICAL MATERIAL AND WIRING PRACTICE Course Code : 313015

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme												Total Marks
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
313015	ELECTRICAL MATERIAL AND WIRING PRACTICE	EMW	SEC	1	-	4	1	6	3	-	-	-	-	-	50	20	25@	10	25	10	100

Total IKS Hrs for Sem. : 0 Hrs

Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination

Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 State safety rules/ standards applicable for the given electrical systems. TLO 1.2 Describe the use of the given tools in the given electrical engineering situation. TLO 1.3 Describe the use of the given safety accessories in the given electrical engineering situation. TLO 1.4 Describe the functions/applications of the given components of wiring.	**Unit - I Wiring Components, Tools and Safety Devices** 1.1 IE rules 1956 (Chapter IV-General safety requirements- No. 29 to 46) 1.2 Applications of Tools used in wiring: Pliers, nose pliers, cutter, screw driver, tester, test lamp, crimping tool, continuity tester, outside micrometer, knife. 1.3 Applications of safety Accessories: hand gloves, helmet, boots, goggles, rubber mats, types of fire extinguishers. 1.4 Components with specifications used in wiring systems: different types of switches (single and double pole), plugs, sockets, DBs, MCBs, MCCBs, RCCBs, holders, wires, cables. (No working only ratings needs to be explained for all these components)	Chalk-Board Presentations Videos Demonstrations, Role play

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Explain the salient features of the given type of conductor with respect to the needed functional properties. TLO 2.2 Explain with justification the applications for the given electrical conductors in specified situations. TLO 2.3 Describe with justification the use of various magnetic materials in the given electrical engineering situation.	**Unit - II Conducting and Electromagnetic Materials** 2.1 Conducting materials: Electrical, Mechanical and Thermal Properties. 2.2 Applications of Conducting materials: copper, aluminium, tungsten, brass, bronze, mercury, silver, lead, nickel and tin. 2.3 Magnetic materials- silicon Steel (CRGO, HRGO) and Amorphous material: properties and their applications	Chalk-Board, Presentations, Demonstration
3	TLO 3.1 Explain the properties of the given electrical insulating materials. TLO 3.2 Classify insulating materials based on working temperature TLO 3.3 Describe the failure phenomena in the given type of insulating material(s). TLO 3.4 Suggest relevant insulating material(s) for the given application(s) with justification.	**Unit - III Electrical Insulating Materials** 3.1 Significance and properties of electrical insulating materials: electrical, mechanical and thermal properties. 3.2 Classification of insulating materials based on working temperature. 3.3 Causes of failure of insulating materials 3.4 Applications of insulating materials in electrical machines and devices.	Chalk-Board, Presentations, Video Demonstration, Model Demonstration

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Explain with justification the criteria for selecting wire/cable and other electrical components for the given type of installation. TLO 4.2 Describe with sketches the installation of wiring systems for the given type of occupancy. TLO 4.3 Describe with sketches the wiring type as per the functional requirements of the given type of occupancy. TLO 4.4 Explain the process of installing the given type of cable(s).	**Unit - IV Electrical Wiring** 4.1 Types of wires and cables, components and accessories of electrical wiring systems. 4.2 Electrical Wiring systems (PVC casing-capping, conduit and concealed), panel wiring 4.3 Electrical Wiring types (one lamp control, staircase and godown) 4.4 Cable laying, Cable joints (terminations), proper size lugs, crimping of joints.	Chalk-Board, Presentations, Demonstration, Videos

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Recommend with justification the necessity of the type of earthing in the given electrical installation system(s). TLO 5.2 Explain the criteria for recommending the earthing system for the given electrical installation. TLO 5.3 Describe with sketches the installation of the given earthing system. TLO 5.4 Describe the testing procedure for the given earthing systems.	**Unit - V Earthing Systems** 5.1 Types of earthing systems (Rod, pipe, plate, chemical earthing). 5.2 Installation of earthing systems. 5.3 Measurement of earthing resistance by Earth tester, Earth resistance values for various installations as per IEEE standards 5.4 Adverse effects of improper earthing system, methods to reduce the earth resistance	Chalk-Board, Presentations, Videos, Model demonstration

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Use different electrical safety accessories and follow safe practices.	1	*Use of different electrical safety accessories and follow safe practices.	2	CO1
LLO 2.1 Douse the class ‘A’ type fire with suitable medium.	2	*Dousing of class ‘A’ type fire with suitable medium.	2	CO1
LLO 3.1 Rescue a person and apply respiratory methods.	3	*Rescue a person and practice artificial respiration.	2	CO1

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 4.1 Use different types of electrical/electronic tools.	4	*Use of different types of electrical/electronic tools.	2	CO1
LLO 5.1 Test the working of single pole one way and two way switches and MCB.	5	Testing of single pole one way, two way switches and MCB using relevant tools and instruments.	2	CO1
LLO 6.1 Operate the MCCB	6	Demonstration of MCCB	2	CO1
LLO 7.1 Test the working of fuse.	7	Testing of rewirable fuse.	2	CO1
LLO 8.1 Prepare series lamp test board with 2 m wire extention.	8	*Preparation of series lamp test board with 2m wire extention.	2	CO1
LLO 9.1 Test the performance of the RCCB.	9	Testing of the RCCB.	2	CO1
LLO 10.1 Choose the appropriate fuse rating and its location for the given circuit.	10	*Selection of fuses in different lighting circuits.	2	CO2
LLO 11.1 Measure insulation resistance of cables using insulation tester	11	*Measurement of insulation resistance of cables using insulation tester	2	CO3
LLO 12.1 Select insulating materials for specific applications.	12	Selection of insulating materials for specific applications from given samples (at least five).	2	CO3
LLO 13.1 Measure insulation resistance of electrical installation using insulation tester	13	*Insulation resistance test on electrical installation.	2	CO3
LLO 14.1 Test insulating oil for its dielectric strength.	14	*Dielectric strength test of given insulating oil sample.	2	CO4
LLO 15.1 Carry out staircase wiring LLO 15.2 Test the working of staircase wiring.	15	Preparation of staircase wiring and its testing.	2	CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 16.1 Carry out godown wiring. LLO 16.2 Test the working of godown wiring.	16	Preparation of godown wiring and its testing.	2	CO4
LLO 17.1 Carry out switch board wiring LLO 17.2 Test the working of switch board.	17	*Preparation of switch board containing four switch, four socket arrangements (with MCB, indicator etc.).	4	CO4
LLO 18.1 Fix and test LED tube.	18	LED tube light mounting, testing and fault finding.	2	CO4
LLO 19.1 Trace cable laying.	19	Power cable tracing. (For machine installation in laboratory)	2	CO4
LLO 20.1 Carry out the polarity test of the electrical installation of machine laboratory.	20	*Electrical installation testing.	2	CO4
LLO 21.1 Draw and trace LT cable.	21	LT cable tracing. (from LT substation-transformer of your college to your laboratory.)	2	CO4
LLO 22.1 Carry out electrical wire joints.	22	*Preparation of electrical wire joints (simple twist, married, Tee and western union joints).	2	CO4
LLO 23.1 Carry out electrical wire joints.	23	*Preparation of electrical wire joints (britannia straight, Britannia tee and rat tail joints).	2	CO4
LLO 24.1 Carry out lug crimping for cable.	24	Lug crimping for cable leads.	2	CO4
LLO 25.1 Carry out PVC casing-capping and conduit wiring.	25	*Preparation of PVC casing-capping, conduit wiring for minimum four points of 3m length.	4	CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 26.1 Carry out wiring to control lamp from different places.	26	One lamp control from three and/or four different places.	2	CO4
LLO 27.1 Trace and draw electrical schematic drawings of a panel.	27	*Tracing of electrical schematic drawings of a panel of any electrical machine in your laboratory.	2	CO4
LLO 28.1 Carry out plate earthing.	28	*Plate earthing.	2	CO5
LLO 29.1 Carry out chemical earthing.	29	Chemical earthing.	2	CO5
LLO 30.1 Test / measure earthing resistance of electrical installation.	30	Testing and measurement of earthing resistance.	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Assignment

- Prepare chart about the information of different types of magnetic materials (ferromagnetic, paramagnetic, diamagnetic).
- Draw the sketches of plate/pipe/rod/chemical earthing.
- Draw the wiring diagram of Flat/Bunglow 1BHK/2BHK.
- Prepare chart about the information of different types of magnetic materials (ferromagnetic, paramagnetic, diamagnetic).
- Draw the wiring diagram of electrical panel.

- Draw symbols related to electrical accessories and wiring.
- Draw the hysteresis loops for hard steel, wrought iron and alloyed steel.
- Identify various parts of electrical panel.

Visit

- Visit to a nearby construction site and observe the electrification work being carried out and note details of wires, switchgears, earthing practices, safety aspects being followed etc.

Micro project

- Collect the information about distribution substation earthing and submit report on it.
- Collect the sample/information about different types and sizes of wires, cables, and switches available in the market and submit report on it.
- Collect information from internet or otherwise on the different electromagnetic materials along with the forms in which they are available and submit report on it.
- Carry out profile lighting upto 5m length with suitable driver (choke).
- Collect the information about methods of wiring and submit report on it.
- Collect the information about MCBs and MCCBs of different specifications and submit report on it.
- Collect the information about RCCBs of different specifications and submit report on it.

Self learning topics

- Latest tools and techniques in the field of electrical wiring, earthing, materials.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Various fuse mounting units, fuse wire of different rating, ammeter, lamp bank.	10
2	Insulation tester 500V or 1000V	11,13
3	Insulating materials of different classes, electric iron	12
4	Dielectric oil test kit, dielectric oil samples	14
5	Wooden/PVC board, two way switches (6A)- 2 Nos, lamp holder- 1 No., lamp- 1 No.	15
6	Wooden/PVC board, one way switch- 1 No., two way switches (6A)- 2 Nos, lamp holder- 3 No., lamp- 3 No.	16
7	Wooden/PVC board, single pole switches (6A)- 4 Nos., sockets (5 pin-6A), MCB- 2A, red color indicator- 1No.	17

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
8	18/20 Watt LED tube with mounting brackets and screws	18
9	Bucket filled with water, sand, Class ‘A’ type fire extinguisher.	2
10	Electric tester, test lamp, multimeter.	20
11	Electrician’s knife, stainless steel rule, diagonal cutting plier, combination plier, wooden mallet, bastard flat file, hard vice, wires of various sizes, bare copper wire, GI wire, sand paper, cotton cloth.	22,23
12	Crimping lugs, crimping tool, combination plier, knife.	24
13	PVC casing capping-3 meter, PVC conduit -3 meter, wires, wooden/PVC board, switches and sockets	25
14	Wooden/PVC board, lamp holder, lamp, 2 way switches(6A)- 2 Nos.	26
15	Copper plate, salt, wood coal, copper or GI wire etc.	28
16	Chemical mixture containing Bentonite, salt, charcoal, chemical electrode, Copper or GI strip/conductor	29
17	Wooden stick, rubber mat, chart or videos of rescue procedure and respiratory methods.	3
18	Earth tester (Anolog/Digital)	30
19	Pliers, screw driver set, nose pliers, measuring tape, cutter cum insulation remover, screw driver, tester, test lamp, crimping tool, lugs, continuity tester, outer micrometer, knife, soldering gun	4
20	Single pole one way and two switches (6A) – 1 No. each, MCB 1/2A- 1No. each, MI/Digital type AC Ammeter 0-10 A, Lamp bank- 10A.	5,7
21	MCCB TP- 100A- 1 No. MCCB TPN- 100A- 1No.	6
22	Rewirable Kitkat fuse, fuse wire.	7
23	Wooden/PVC board, lamp holder, lamp, extention wire 2m	8
24	RCCB- 16 A double pole, sensitivity 30mA, lamp bank, switch.	9

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
25	Safety hand gloves, safety boots, safety goggles, safety rubber mats, safety helmet (All ISI Mark).	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Wiring Components, Tools and Safety Devices	CO1	3	0	0	0	0
2	II	Conducting and Electromagnetic Materials	CO2	3	0	0	0	0
3	III	Electrical Insulating Materials	CO3	3	0	0	0	0
4	IV	Electrical Wiring	CO4	3	0	0	0	0
5	V	Earthing Systems	CO5	3	0	0	0	0
Grand Total				**15**	**0**	**0**	**0**	**0**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- For formative assessment of laboratory learning 50 marks. Each practical will be assessed considering appropriate % weightage to process and product and other instructions of assessment.

Summative Assessment (Assessment of Learning)

- End semester summative assessment of 25 marks for laboratory learning.

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3	2	2	3	3	1	3			
CO2	3	2	3	1	1	1	2			
CO3	3	2	3	1	1	1	2			
CO4	3	3	2	3	1	2	2			
CO5	3	1	1	2	2	2	2			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Bhattacharya S. K.	Electrical Engineering Drawing	New Age International, New Delhi, ISBN: 978-81-224-0855-3.

Sr.No	Author	Title	Publisher with ISBN Number
2	Uppal S.L; Garg G.C.	Electrical Wiring, Estimating and Costing	Khanna Publishers, New Delhi, ISBN-13: 978-81-7409-240-3.
3	Singh R.P.	Electrical Workshop: Safety, commissioning, maintenance and testing of electrical equipment	I.K. International Publishing House , Pvt. Ltd. New Delhi, ISBN:978-9389447057
4	Gupta J. B.	Electrical Estimating and Costing	S. K. Kataria & Sons, New Delhi, ISBN:978-93-5014-279-0
5	Indulkar C.S. & Thiruvengadam S.	An introduction to electrical engineering materials	S Chand & Co., ISBN :978-8121906661

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	www.bharatskills.gov.in	Directorate general of training - central repository for skills in NSQF curriculum
2	https://www.osha.gov/sites/default/files/publications/osha3075.pdf	Controlling Electrical Hazards
3	https://nsc.org.in/	National safety council of India
4	https://www.esfi.org/	Electrical safety foundation

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

Programme Name/s	**: Electrical Engineering/ Electrical Power System**
Programme Code	**: EE/ EP**
Semester	**: Fourth**
Course Title	**: COMPUTER AIDED DRAWING AND SIMULATION**
Course Code	**: 314008**

I. RATIONALE

It is the need of the industry to draw electrical engineering drawings and use CAD software effectively as per the requirement. In this course, students will practice to interpret drawings, communicate ideas, and turn concepts into practical designs. They gain skills in navigating CAD software and using its tools efficiently to draw electrical drawings. This course is designed in such a way that practical performed in this course will enhance their skills to compete in fast growing electrical industry and understand different circuits by simulation.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Draw electrical drawings using CAD and simulate basic Electrical circuits using simulation software(s).

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Comprehend Electrical Drawings.
- CO2 - Locate various components of CAD software.
- CO3 - Use relevant CAD Tools and Commands for Electrical Drawings.
- CO4 - Draw different Electrical Drawings using CAD software.
- CO5 - Simulate Basic Electrical and Electronic circuits.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

COMPUTER AIDED DRAWING AND SIMULATION **Course Code : 314008**

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
314008	COMPUTER AIDED DRAWING AND SIMULATION	CDS	SEC	-	-	4	-	4	2	-	-	-	-	-	25	10	25@	10	-	-	50

Total IKS Hrs for Sem. : 0 Hrs

Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination

Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Sketch the electrical symbols as per requirement in electrical drawings. TLO 1.2 Interpret given electrical power/control wiring diagram. TLO 1.3 Identify types of electrical panel. TLO 1.4 Sketch GA Diagram of Electrical control panel (Assume suitable dimensions).	**Unit - I Electrical Drawings.** 1.1 Symbols: Electrical and Electronic as per SP 30: 2023 Part 1, section 3. 1.2 Types of electrical drawings (a) Power wiring diagram (single line diagram (SLD) or Multiline diagram) (b) Control wiring diagram (Schematic diagram) (c) Block diagrams (d) Pictorial diagrams. 1.3 Types of Electrical panels (a) MCC (Motor control center) Panel (b) PCC (Power control center) panel (c) APFC (Automatic Power Factor Controller) Panel (d) PLC (Programmable logic controller) Panel. 1.4 General Arrangement (GA) diagram of Electrical control panel.	Hands-on Presentations Lecture Using Chalk-Board
2	TLO 2.1 Identify the function of the given components of CAD classic screen. TLO 2.2 Identify the given components of CAD screen. TLO 2.3 Identify the given toolbar and commands.	**Unit - II Computer Aided Design (CAD) Introduction.** 2.1 Components of CAD classic screen. 2.2 Menu bar and status bar. 2.3 Open and Save file. 2.4 CAD Toolbars. 2.5 Command Box. 2.6 Zoom in and Zoom out.	Hands-on Presentations Demonstration

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Use the coordinate methods and practice basic commands. TLO 3.2 Practice Draw, Modify and Annotation toolbar commands. TLO 3.3 Practice Important CAD Modes.	**Unit - III CAD Tools and Commands.** 3.1 Coordinate Method: Absolute, Relative, Polar. Basic commands: Limits, Units. 3.2 Draw Toolbar: Line, Polyline, Circle, Arc, Rectangle, Ellipse, Polygon, Hatch. 3.3 Modify Toolbar: Move, Rotate, Trim, Erase, Copy, Cut, Mirror, Fillet, Chamfer, Offset, Explode, Strech, Scale. 3.4 Annotation Toolbar: Multiline Text, Single Line Test, Linear dimension, Aligned dimension, Angular Dimension, Arc Length Dimension, Radius Dimension, Diameter Dimension 3.5 Important CAD Modes : Grid, Ortho, Snap, Polar Tracking, Object Snap Tracking.	Hands-on Demonstration Presentations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Sketch the power wiring diagram, control wiring diagram and GA Diagram of Electrical control panel using CAD TLO 4.2 Sketch the Single line diagram (SLD) of the 11 kV/433 V distribution substation using CAD TLO 4.3 Sketch the Single line diagram (SLD) of residential/commercial unit using CAD TLO 4.4 Sketch the Single line diagram (SLD) of any industrial plant using CAD	**Unit - IV Use of CAD in Real World Electrical Engineering Drawings.** 4.1 Applications of electrical CAD software to: (a)Draw power wiring diagram of electrical control panel. (b)Draw control wiring diagram of electrical control panel. (c)Draw GA diagram of electrical control panel. 4.2 Applications of electrical CAD software to Single line diagram (SLD) of the 11 kV/433 V distribution substation. 4.3 Prepare Single line diagram (SLD) of residential/commercial unit using CAD. 4.4 Draw the Single line diagram (SLD) of any industrial electrical installation using CAD.	Hands-on Demonstration Presentations

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Select and use softwares for Electrical and electronic circuit simulations. TLO 5.2 Build, Simulate and Test Basic electric circuits. TLO 5.3 Build, Simulate and Test Basic electronic circuits. TLO 5.4 Measure various electrical parameters and Generate or plot relevant Waveforms/Graphs. TLO 5.5 Develop P.C.B. layout for a given electrical circuit using software.	**Unit - V Simulation of Electrical and Electronic Circuits.** 5.1 Voltage, current, power across (a)Series R-L circuit (b)Series R-C circuit (c)Series R-L-C circuit. 5.2 Rectifier circuit, KVL and KCL simulation. 5.3 Triac Lamp Dimmer Circuit simulation. 5.4 Basic Logic Gate and adder circuit simulation. 5.5 Printed Circuit Board (PCB) preparation basic information.	Hands-on Demonstration Presentations

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Draw symbols of different electrical and electronic components using drawing instruments.	1	*Symbols of Electrical and Electronic Components as per SP 30: 2011(NEC 2011) part 1, section 3 or new equivalent IS on sketch book.	2	CO1
LLO 2.1 Draw Power and control wiring diagram for DOL starter.	2	*Power and Control wiring diagram of DOL Starter on sketch book.	2	CO1
LLO 3.1 Draw Power and control wiring diagram for Star Delta starter.	3	*Power and Control wiring diagram of Star Delta Starter on sketch book.	2	CO1

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 4.1 Draw General Arrangement (GA) Diagram for DOL/Star delta starter panel or any other electrical panel.	4	General Arrangement Diagram for Electrical Panel on sketch book (Assume suitable dimensions).	2	CO1
LLO 5.1 Install CAD software. LLO 5.2 Create new drawing. LLO 5.3 Locate components of CAD Classic Screen (CAD screen layout, Drawing area, Menu and Toolbars, Status bar).	5	*Different components of CAD classic screen.	2	CO2
LLO 6.1 Create and Save drawing. LLO 6.2 Set the drawing Limits and Units of the file. LLO 6.3 Perform Zoom in and Zoom out functionality.	6	*CAD file operations and Limits & Units of Drawing.	2	CO2
LLO 7.1 Use Draw Toolbar of CAD for drawing basic geometrical shapes.	7	* Basic geometrical shapes using Draw Toolbar commands (Line, Polyline, Circle, Arc, Rectangle, Ellipse, Polygon, Hatch).	2	CO2 CO3
LLO 8.1 Use Modify Toolbar of CAD for modifying or editing CAD drawing.	8	Modifying or editing basic geometrical shapes using modify commands (Move, Rotate, Trim, Erase, Copy, Cut, Mirror, Fillet, Chamfer, Offset, Explode, Stretch, Scale).	2	CO2 CO3

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 9.1 Use Annotation Toolbar of CAD for writing Text and measuring dimensions.	9	*Annotation Toolbar commands (Multiline Text, Single Line Test, Linear dimension, Aligned dimension, Angular Dimension, Arc Length Dimension, Radius Dimension, Diameter Dimension).	2	CO2 CO3
LLO 10.1 Use Important CAD modes for drawing: Grid, Ortho, Snap, Polar Tracking, Object Snap Tracking.	10	*Important CAD Modes for drawing: Grid, Ortho, Snap, Polar Tracking, Object Snap Tracking.	2	CO2 CO3
LLO 11.1 Draw symbols of different electrical and electronic components using CAD.	11	*Symbols of Electrical and Electronic Components as per SP 30: 2011(NEC 2011) part 1, section 3 or new equivalent IS using CAD.	2	CO1 CO2 CO3
LLO 12.1 Draw Power and control wiring diagram for DOL starter using CAD.	12	*Power and Control wiring diagram of DOL Starter using CAD.	2	CO2 CO3
LLO 13.1 Draw Power and control wiring diagram for Star Delta starter using CAD.	13	*Power and Control wiring diagram of Star Delta Starter using CAD.	2	CO2 CO3
LLO 14.1 Draw General Arrangement (GA) Diagram for DOL/Star delta starter panel or any other electrical panel using CAD.	14	*General Arrangement Diagram for Electrical Panel (Assume suitable dimensions) using CAD.	2	CO2 CO3
LLO 15.1 Draw Single Line Diagram (SLD) of the 11kV/433V distribution substation using CAD software.	15	*Single Line Diagram (SLD) of the 11kV/433V distribution substation using CAD.	2	CO2 CO3 CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 16.1 Draw Single Line Diagram (SLD) of any Industrial Electrical Installation using CAD Software. LLO 16.2 Plot and print drawings to produce hard copies or digital outputs.	16	*Single Line Diagram (SLD) of any Industrial Electrical Installation using CAD Part I.	2	CO2 CO3 CO4
LLO 17.1 Draw Single Line Diagram (SLD) of any Industrial Electrical Installation using CAD Software. LLO 17.2 Plot and print drawings to produce hard copies or digital outputs.	17	*Single Line Diagram (SLD) of any Industrial Electrical Installation using CAD Part II.	2	CO2 CO3 CO4
LLO 18.1 Install simulation software. LLO 18.2 Create new simulation worksheet. LLO 18.3 Use different tools available in software.	18	*Use of simulation software.	2	CO5
LLO 19.1 Build ohms law, series & parallel circuit using simulation software. LLO 19.2 Measure different electrical parameters using software tools.	19	*Simulation for verification of Ohm's law and series & parallel resistances in circuit.	2	CO5

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 20.1 Build KCL and KVL Circuit using software. LLO 20.2 Measure electrical parameters using software.	20	*Simulation of Kirchoff's Current Law and Kirchoff's Voltage Law.	2	CO5
LLO 21.1 Build R-L series circuit using software. LLO 21.2 Measure electrical parameters using software. LLO 21.3 Observe Relevant waveforms across each components.	21	*Simulation of R-L series circuit.	2	CO5
LLO 22.1 Build R-C series circuit using software. LLO 22.2 Measure electrical parameters using software . LLO 22.3 Observe Relevant waveforms across each components.	22	Simulation of R-C series circuit.	2	CO5
LLO 23.1 Build PN junction diode circuit using software. LLO 23.2 Observe Diode characteristics.	23	Simulation of VI Characteristics of diode.	2	CO5

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 24.1 Build half-wave rectifier circuit using software. LLO 24.2 Measure different parameters using software. LLO 24.3 Develop P.C.B. layout for a given electrical circuit using software.	24	*Simulation of single phase half-wave rectifier circuit.	2	CO5
LLO 25.1 Build full-wave rectifier circuit using software. LLO 25.2 Measure different parameters using software. LLO 25.3 Develop P.C.B. layout for a given electrical circuit using software.	25	Simulation of single phase full-wave rectifier circuit.	2	CO5
LLO 26.1 Build basic logic gates circuit using software. LLO 26.2 Observe different parameters using software.	26	*Simulation of Basic Logic Gates.	2	CO5
LLO 27.1 Build Triac Lamp Dimmer circuit using software. LLO 27.2 Observe different parameters using software.	27	*Simulation of Triac Lamp Dimmer circuit.	2	CO5

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 28.1 Build Half and Full Adder Logic circuit using software. LLO 28.2 Observe different parameters using software. LLO 28.3 Develop P.C.B. layout for a given electrical circuit using software.	28	Simulation of Half and Full Adder Logic circuit.	2	CO5
LLO 29.1 Build Half and Full Subtractor circuit using software. LLO 29.2 Observe different parameters using software.	29	Simulation of Half and Full Subtractor circuit.	2	CO5
LLO 30.1 Build any circuit using software. LLO 30.2 Develop P.C.B. layout for a given electrical circuit using software.	30	P.C.B. Layout Preparation for electrical circuit using software.	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Assignment

- Simulate stair case wiring circuit
- Simulate one switch one bulb house wiring diagram circuit
- Simulate Op-Amp integrator circuit design
- Simulate & Measure average power and power factor with a wattmeter
- Simulate series and parallel RLC circuit
- Study EPLAN software

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	A4 Sketch Book Drawing Material	1,2,3,4

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
2	Simulation Software List 1) Any Open-Source Software like Scilab. 2) Multisim Educational Version 14.3 3) PSIM 11.1	18,19,20,21,22,23,24,25,26,27,28,29,30
3	CAD Software List 1) Any Open-Source Computer Aided Design (CAD) Software. 2) LibreCAD. 3) AutoCAD Electrical Student Version.	5,6,7,8,9,10,11,12,13,14,15,16,17
4	Computer Sysytem Operating System: 64-bit Windows 8 or higher Processor: 2.5–2.9 Ghz processor / Recommended: 3+ Ghz processor RAM: 8 GB as a minimum, with 16GB being recommended GPU: 1GB of VRAM as a minimum with DirectX 11 support; Recommended: 4 GB of VRAM with DirectX 12 support Storage: 10 GB.	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table) : NOT APPLICABLE

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Teacher should prepare rubrics for Formative assessment
- Each Practical will be assessed for 25 Marks and average of all marks obtained will be considered.

Summative Assessment (Assessment of Learning)

- End Semester assessment of 25 marks for laboratory learning.
- Teacher should prepare rubrics for Summative Assessment.

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3	1	1	3	2	-	2			
CO2	3	-	-	3	-	-	2			
CO3	3	-	2	3	1	-	2			
CO4	3	1	3	3	1	2	2			
CO5	3	1	3	3	1	2	2			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Cornel Barbu	Electrician's Book how to Read Electrical Drawings	Lulu.com, ISBN-13: 9781435713208
2	Prof. Sham Tickoo	AutoCAD Electrical 2021: A Tutorial Approach, 2nd Edition	CADCIM Technologies, ISBN-13 9781640571006, 1640571000
3	John Reeder, Reeder	Using Multisim 9 Troubleshooting DC/AC Circuits	Delmar Cengage Learning, ISBN-13 9781111322137, 1111322139

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.kicad.org/	Kicad : This link download Open Source PCB Design Kicad Software
2	https://www.autodesk.com/education/students	AutoCAD : Register and get free student version of LATEST AutoCAD software
3	https://law.resource.org/pub/in/bis/S05/is.sp.30.2011.pdf	This link downloads IS SP:30 2011 (NEC 2011)
4	https://powersimtech.com/products/	PSIM : This link downloads PSIM software demo version
5	https://powersimtech.com/wp-content/uploads/2021/01/PSIM-Use r-Manual.pdf	This link downloads PSIM software user Manual
6	https://scilab.in/DownloadScilab	Scilab : This link downloads Scilab software
7	https://librecad.org/	LibreCAD : This link downloads Open Source LibreCAD software

Sr.No	Link / Portal	Description
8	https://www.falstad.com/circuit/	Falstad : This is an electronics circuit simulator applet
9	https://www.ni.com/en/support/downloads/software-products/do wnload.multisim.html#452133	NI Multisim : This is an electrical and electronics circuit simulator
10	https://www.youtube.com/watch?v=GH-JFXbOcZg&t=71s	Hartley Oscillator circuit simulation on Multisim software
11	https://www.youtube.com/watch?v=mzglU-tMgXY	Simulating halfwave and full wave rectifier circuit in multisim
12	https://www.youtube.com/watch?v=szfgbN0GD5A	AutoCAD practice exercise
13	https://www.youtube.com/watch?v=_2d_Tb9bzsQ&t=10s	Series RLC Circuit Simulation using Multisim
14	https://www.youtube.com/watch?v=UKpIGwto47U	Triac Lamp Dimmer Circuit
15	https://youtu.be/9m8ABCSKTec?si=Kuf6ryURVs9hpK49	VI Characteristics of PN junction diode 1N4007

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students